

Yash Patel

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EDUCATION

University of New Hampshire

Bachelor of Science in Computer Engineering

Durham, New Hampshire

May 2026

- **Courses:** FPGA Design, Embedded Systems Design, Digital Systems Design, Electronic Circuit Design, Communication Systems Design, Operating Systems, Computer Architecture

EXPERIENCE

University of New Hampshire — InterOperability Labs

Senior Test Engineer

Durham, New Hampshire

Apr 2025 - Present

- **2.5/5/10GBASE-T1 Ethernet Decoder:** Engineered the first commercially available multi-gigabit Single-Pair Ethernet decoder and analysis tool in Rust, recovering PAM-4 symbols, decoding infobits, and extracting data from RS-FEC protected frames from raw oscilloscope captures at 60x the throughput of the legacy MATLAB based system.
- **MultiGBASE-T1 SPE Tester:** Validated multi-gigabit Single-Pair Ethernet PHY-Control (IEEE 802.3 Clause 149) implementations, analyzing NRZ/PAM-4 signal integrity, RS-FEC performance, and link training state machines using an automated Rust and Verilog platform
- **Influenced future ISO 21111-6 test plan revisions:** Delivered technical feedback, test expertise, and critical fixes to OPEN Alliance, driving revisions progressing toward formal ISO standards.

Junior Test Engineer

Jan 2024 - Apr 2025

- **Development for custom FPGA test automation:** Architected an IEEE 802.3 Clause 22/45 MDIO transmitter in Verilog for a custom FPGA test automation framework, integrated into commercial-grade hardware validation suites.
- **Single-Pair Ethernet automation testing:** Built a 100BASE-T1 PHY Control and PCS automation tool in Rust and MATLAB that ingests oscilloscope captures over SCPI and programmatically determines pass/fail, removing manual analysis from the test loop.
- **Reservation Manager and Tester for 20+ bookings:** Managed the end-to-end testing lifecycle for concurrent vendor bookings, delivering detailed testing reports and analyzing Single-Pair Ethernet (SPE) transmissions.

INDUSTRY CONTRIBUTIONS

- **Acknowledged Contributor, OPEN Alliance IEEE 100BASE-T1 Test Suite V1.4:** Recognized in the specification acknowledgments for driving major technical revisions, including optimizing test coverage, deprecating obsolete tests, and refining functional requirements for conformance testing.

SKILLS

- **Languages:** Rust, C, Verilog, MATLAB, Python, HTML, CSS, JavaScript, Bash, Lua
- **Protocols:** I2C, SPI, UART, CAN-BUS, 100BASE-T1 (PHY-C, PCS), TCP/IP, Ethernet, SSH
- **Tools:** Wireshark, Oscilloscopes, Logic Analyzers, Packet Generators, Vivado, Git
- **Platforms & OS:** GNU/Linux, Windows, AWS IoT, AWS S3, STM32, Xilinx FPGA, Altera FPGA

PROJECTS

Smart Adaptive Audio Equalizer Headset

Senior Design Project

- **Adaptive headset architecture:** Collaborated with a team of 3 on a design featuring active external (ambient) and internal (biological) noise cancellation.
- **Digital Signal Processing (DSP) calibration:** Developed a routine using a 60-second automated hearing test to create personalized audio profiles, dynamically mapped over audio sources.
- **Dual-microphone feedback/feedforward system:** Engineered the signal path to eliminate city noise and internal interference (e.g., blood flow).
- **MSP430 Morse Code Decoder:** Programmed an MSP430FR6989 to perform real-time text-to-Morse decoding, utilizing UART for input and GPIO-controlled LEDs and buzzers for signal output.